

A sharp target-atom budget at the critical boundary for Brenier stability

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Abstract

For the fixed critical boundary source of manuscript 008, we determine the joint dependence of the moment-controlled Brenier-map modulus on target distance and the number of target atoms. With $p > 2$, $s = p/(p - 2)$, at most $N \geq 2$ atoms in each target, and unit p th moments, the modulus is comparable to $w^{1/3} \min\{N, \log(e/w)\}^{1/(3s)}$, uniformly in both parameters for sufficiently small w . Thus each fixed finite atom budget restores a pure one-third estimate, with optimal constant growth $N^{1/(3s)}$. A weak- L^s overlap estimate and finite-level pairing give the upper bound. The lower bound tracks the atom count in the previously proved globally mass-matched construction; the two-atom case is treated separately. The potential estimate is an explicit upstream input.

Status and attribution. Written research theorem, prepared with AI assistance. The longer candidate proof received a separate model-conducted mathematical review; this note is not a Lean formalization, external human peer review, or novelty determination. The source, potential theorem, rare-cell mechanism and multiscale convex construction are inherited from manuscripts 001 and 008. The contribution here is the weak-overlap finite-budget bound and the jointly uniform cardinality/distance crossover, not a new proof of those upstream results.

1 Statement and fixed source

Fix $d \geq 2$, $p > 2$, and put

$$s = \frac{p}{p-2}, \quad \beta = s - 1 = \frac{2}{p-2}, \quad \alpha = \frac{1}{3s}.$$

Throughout the source is the single normalized probability

$$r(x) = f_\beta(x_1) \prod_{j=2}^d g(x_j), \quad f_\beta(t) = c_\beta t^\beta e^{-t^2/2} \mathbf{1}_{t>0}, \quad g(z) = c_g e^{-1/(1-z^2)} \mathbf{1}_{|z|<1}, \quad \rho = r \, dx, \quad (1)$$

where each factor integrates to one. Its density is zero off $D = (0, \infty) \times (-1, 1)^{d-1}$. This is precisely the critical source in 008. It has all moments and is 1-strongly log-concave, since on its support

$$(-\log f_\beta)'' = 1 + \beta/t^2, \quad (-\log g)'' = 2(1 + 3z^2)/(1 - z^2)^3 \geq 2.$$

Write T_μ for the quadratic Brenier map from ρ to μ . For each positive integer N , define

$$\mathcal{C}_{p,N} = \{\mu \text{ a probability} : |\text{supp } \mu| \leq N, \int |y|^p \, d\mu \leq 1\},$$

$$\Omega_N(w) = \sup_{\substack{\mu, \nu \in \mathcal{C}_{p,N} \\ W_2(\mu, \nu) \leq w}} \|T_\mu - T_\nu\|_{L^2(\rho)}.$$

Support counts distinct positive-mass atoms; coincident labels are merged and zero weights discarded. Atom locations and weights may both vary, with no lower weight bound or common fixed support.

Theorem 1.1 (Joint atom-budget and distance law). *There exist $c, C > 0$ and $w_0 \in (0, 1)$, depending only on p, d and the fixed normalized source (1), such that for every integer $N \geq 2$ and every $0 < w \leq w_0$,*

$$cw^{1/3} \min\{N, \log(e/w)\}^\alpha \leq \Omega_N(w) \leq Cw^{1/3} \min\{N, \log(e/w)\}^\alpha. \quad (2)$$

All three constants are independent of N . For $N = 1$ one has exactly $\Omega_1(w) = \min\{w, 2\}$ for every $w \geq 0$.

Upstream potential premise. The upper bound uses centered convex Brenier potentials satisfying

$$\|U_\mu - U_\nu\|_{L^2(\rho)} \leq K_\rho W_2(\mu, \nu) \quad (\mu, \nu \in \mathcal{P}_2). \quad (\text{P})$$

Their convex representatives are finite on D . Manuscript 001 v3, Theorem 1.1, supplies $K_\rho = \sqrt{2}$ for (1). This is an explicitly imported written theorem, not independently reproved here. Finite target supports suffice in the present application and give finite-slope potentials. The lower bounds below do not use (P).

2 Weak overlap and a finite-level energy estimate

For a nonnegative random variable a on a probability space put $\|a\|_{s,\infty} = \sup_{t>0} t \Pr(a > t)^{1/s}$, and write $s' = s/(s-1) = p/2$.

Lemma 2.1 (Finite-level pairing). *If $\|a\|_{s,\infty} \leq H$ and $b \geq 0$ takes at most N values, then*

$$\mathbb{E}[ab] \leq s' H N^{1/s} \|b\|_{s'}.$$

More precisely, if the values of b are b_i on disjoint sets of masses m_i , then the right side may be replaced by $s' H \sum_i b_i m_i^{1/s'}$.

Proof. Layer cake gives, for a set E of mass m ,

$$\int_E a \leq \int_0^\infty \min\{m, (H/t)^s\} dt = s' H m^{1/s'}.$$

Sum over the level sets and apply finite-sequence Hölder to $b_i m_i^{1/s'}$ and the sequence of ones. Zero cases are immediate. No minimum m_i is needed. $\square$

For $h > 0$, define on $\{r > 0\}$

$$d_{j,\pm h} = (1 - r(x \pm h e_j)/r(x))_+, \quad W_{j,h} = 6|d_{j,h} - d_{j,-h}| + 2d_{j,h}, \quad F_h = \max_j W_{j,h},$$

and set these functions to zero on $\{r = 0\}$. Always $0 \leq F_h \leq 8$.

Lemma 2.2 (Critical weak overlap). *For the fixed source (1) there is $B < \infty$ such that, for $0 < h \leq 1/4$,*

$$\|F_h\|_{L^{s,\infty}(\rho)} \leq Bh. \quad (3)$$

Consequently, for sufficiently small h , $\|F_h\|_{L^s(\rho)} \leq Ch(1 + \log(1/h))^{1/s}$.

Proof. Product structure reduces each relative loss to its own marginal. Integrating the logarithmic derivative $\beta/x - x$ when $x > 2h$, and using the bound one when $x \leq 2h$, gives

$$d_{1,\pm h} \leq \min\{1, C_\beta h(X + 1 + X^{-1})\}.$$

Here $X \sim f_\beta$, $\Pr(X < a) \leq Ca^{\beta+1} = Ca^s$ for all $a > 0$, and $\mathbb{E}X^s < \infty$. A union bound and Markov's inequality show that $X + 1 + X^{-1}$ has bounded weak- L^s norm.

For a transverse factor put $a = 1 - |z|$. If $a > 2h$, its logarithmic derivative on the translated segment has magnitude at most C/a^2 . If $a \leq 2h$ and $h \leq 1/4$, then $h/a^2 \geq 1$. Hence

$$d_{j,\pm h} \leq \min\{1, Ch(1 - |Z_j|)^{-2}\}.$$

All moments of $(1 - |Z_j|)^{-2}$ under g are finite. Use Markov and $W_{j,h} \leq 14 \max_{\pm} d_{j,\pm h}$, and then take a finite union over coordinates and signs. This proves (3), including boundary losses from the zero extension. Finally integrate $s \int_0^8 t^{s-1} \rho(F_h > t) dt$ and split at Bh to obtain the strong bound. $\square$

3 The two upper estimates

For convex u, v finite on D , put $\delta = \inf_a \|u - v - a\|_{L^2(\rho)}$ and $X = \|\nabla u - \nabla v\|_{L^2(\rho)}$. The exact minimum-weight estimate is

$$X^2 \leq 12d\delta^2 h^{-2} + \int F_h(|\nabla u|^2 + |\nabla v|^2) d\rho. \quad (4)$$

We recall its short proof to specify the domain and coefficients. Fix a coordinate, let $m_h = \min\{r(x), r(x + he_j)\}$, and write $z = \partial_j u$. On $\{m_h > 0\}$, convexity gives $z(x) \leq D_h u(x) \leq z(x + he_j)$. The signed-square inequality $(b - a)^2 \leq 2(b|b| - a|a|)$ for $a \leq b$, followed by translation, yields

$$\int |D_h u - z|^2 m_h dx \leq 2 \int |z|^2 r |d_{j,h} - d_{j,-h}| dx.$$

The weights after translation are at most r , so all terms are integrable. Also $\int |D_h(u - v - a)|^2 m_h \leq 4h^{-2} \|u - v - a\|_2^2$. The three-term square inequality gives coefficient six in the individual gradient energies. On the missing weight $r - m_h = rd_{j,h}$, the inequality $|a - b|^2 \leq 2(|a|^2 + |b|^2)$ gives coefficient two. Sum over coordinates and infimize in a . Convexity of D keeps every segment with $m_h > 0$ in the domain. This proves (4).

Now take transport potentials for $\mu, \nu \in \mathcal{C}_{p,N}$ and $W_2(\mu, \nu) \leq w$. Each energy $|T_\mu|^2$ takes at most N values and has $L^{s'}$ norm at most one. By (P), Lemmas 2.1–2.2,

$$\|T_\mu - T_\nu\|_2^2 \leq 12dK_\rho^2 w^2 h^{-2} + 2s' BhN^{1/s}.$$

Set $h = w^{2/3} N^{-1/(3s)}$, which is at most $1/4$ for sufficiently small w independently of N . This gives

$$\Omega_N(w) \leq Cw^{1/3} N^{1/(3s)}. \quad (5)$$

Alternatively use strong Hölder and the last assertion of Lemma 2.2 in (4). Set $L = \log(e/w)$ and $h = w^{2/3}L^{-1/(3s)}$. Then $\log(1/h) \leq C_s L$ and

$$\Omega_N(w) \leq Cw^{1/3}L^{1/(3s)}, \quad (6)$$

without needing finite support. The minimum of (5) and (6) proves the upper half of (2).

Remark 3.1. The finite-budget upper bound holds at any source satisfying (P) and (3), with the relevant domain hypotheses. More precisely, the target contribution may be bounded by $s'Bh \sum_i m_i^{2/p} |y_i|^2$ instead of $s'BhN^{1/s}$. The weak/finite-level pairing is elementary; no first-use claim is made for Lorentz-space reasoning.

4 Lower bounds with the atom count retained

The construction in the next lemma is inherited from 008, Section 7. Its proof is recalled with the exact cardinality and uniformity needed here; it is not counted as a new construction.

Lemma 4.1 (Uniform finite-strip family). *There are constants $c_1, C_1, \varepsilon_0 > 0$, independent of K, h , such that for every integer $K \geq 1$ and $0 < h \leq \varepsilon_0 2^{-K}$ there are targets $\mu_{K,h}, \nu_{K,h} \in \mathcal{C}_{p,2K+1}$ with*

$$\|T_{\mu_{K,h}} - T_{\nu_{K,h}}\|_2 \geq c_1 K^{1/(2s)} h^{1/2}, \quad W_2(\mu_{K,h}, \nu_{K,h}) \leq C_1 K^{1/(2s)} h^{3/2}. \quad (7)$$

Proof. Let X, Z have independent laws f_β, g , and let $F(t) = \Pr(X < t)$, zero for $t \leq 0$. Fix $k = 1/4$ and set $t_n = 2^{-n}$, $R_n = t_n^{-\beta/2}$ for $1 \leq n \leq K$. Take $\varepsilon_0 \leq 1/12$ initially, so the windows $[t_n - 3h, t_n + 3h]$ are positive and disjoint. For $H_h(y) = \mathbb{E}F(y + khZ)$ define

$$\ell_n = H_h^{-1}(F(t_n - h)), \quad r_n = H_h^{-1}(F(t_n + h)).$$

Strict monotonicity supplies these inverses on the relevant positive interval. Bracketing by $F(y \pm kh)$ gives $|\ell_n - (t_n - h)|, |r_n - (t_n + h)| \leq kh$. Since $\mathbb{E}Z = 0$, Taylor expansion and the mean value theorem improve these to $O(h^2/t_n)$. Uniformly on the window this uses $f_\beta(y) \asymp f_\beta(t_n)$, $|f'_\beta(y)| \leq C f_\beta(t_n)/t_n$, and $H'_h(y) \geq c f_\beta(t_n)$.

Put $d_n = (\ell_n + r_n)/2$, $\gamma_n = (r_n - \ell_n)/(2h)$ and

$$A_n = \{t_n - h < X < t_n + h\}, \quad C_n = \{\ell_n - khZ < X < r_n + khZ\}.$$

The perturbed half-width lies between $h/2$ and $3h/2$. Symmetry gives the two separate exact cumulative identities

$$\mathbb{E}F(\ell_n - khZ) = F(t_n - h), \quad \mathbb{E}F(r_n + khZ) = F(t_n + h). \quad (8)$$

Thus $m_n = \rho(A_n) = \rho(C_n)$. For $D_n = \rho(A_n \triangle C_n)$, the overlapping intervals and the endpoint estimates give

$$m_n \asymp f_\beta(t_n)h, \quad D_n = 2kh f_\beta(t_n) \mathbb{E}|Z| + O(f_\beta(t_n)h^2/t_n).$$

Decrease ε_0 once so $D_n \asymp f_\beta(t_n)h$ uniformly. All constants are independent of K .

For global convex realization, set

$$\Phi(x) = \sum_n 2R_n(t_n - x)_+, \quad \tilde{\Phi}(x) = \sum_n 2R_n(d_n - x)_+.$$

They have the same ordered outer slopes. Let q_n be the average of the slopes adjacent to cusp n , and let $L_n, \tilde{L}_n$ be the supporting lines with slope q_n at t_n, d_n . Between adjacent cusps $\Phi - L_n = R_n|x - t_n|$, and globally the left side is at least the right; similarly for the tilded functions. Define

$$\begin{aligned} U_h(x) &= \max\{\Phi(x_1), L_n(x_1) + R_n h : 1 \leq n \leq K\}, \\ V_h(x) &= \max\{\tilde{\Phi}(x_1), \tilde{L}_n(x_1) + R_n h \gamma_n + R_n h k x_2 : 1 \leq n \leq K\}. \end{aligned}$$

The central planes are active exactly on A_n, C_n on the source; supporting-line bounds and window separation exclude all other interference. There are $K + 1$ outer labels and K central labels, hence $2K + 1$ atoms in each target. All have positive mass and are distinct, though only the upper count is needed. The outer labels agree between the two maps, and the central labels are $q_n e_1$ and $q_n e_1 + R_n h k e_2$.

Both endpoint identities (8) preserve every outer mass as well as each central mass: an intermediate outer mass is the difference of its two bordering cumulative masses. Consequently same-label target coupling is admissible. The horizontal discrepancy on $A_n \triangle C_n$ is exactly R_n , and the vertical discrepancy is $R_n h k$ on C_n . There is no direct left-outer to right-outer jump because the central intervals overlap. Thus

$$\begin{aligned} \|\nabla U_h - \nabla V_h\|_2^2 &= \sum_n R_n^2 D_n + k^2 h^2 \sum_n R_n^2 m_n, \\ W_2((\nabla U_h)_\# \rho, (\nabla V_h)_\# \rho)^2 &\leq k^2 h^2 \sum_n R_n^2 m_n. \end{aligned}$$

Convexity identifies the source maps as Brenier maps; each has finite slopes for fixed K . No optimality of the same-label target coupling is needed.

Geometric summation of the jumps bounds both gradient magnitudes by $C \min\{x_1^{-\beta/2}, t_K^{-\beta/2}\}$ on $0 < x_1 < 1$, with zero slopes for $x_1 \geq 1$. Cusp shifts and vertical slopes change only C . The critical identity $\beta p/2 = \beta + 1$ gives

$$\int (|\nabla U_h|^p + |\nabla V_h|^p) d\rho \leq C_0(K + 1) :$$

the capped part below t_K is bounded and the remainder is at most $C \int_{t_K}^1 dx/x$. Scale both potentials by $\lambda_K = [C_0(K + 1)]^{-1/p}$, so each target moment is at most one. Finally $R_n^2 f_\beta(t_n) = c_\beta e^{-t_n^2/2} \asymp 1$, and $1 - 2/p = 1/s$. The preceding identities give (7) with constants uniform in both K, h . $\square$

4.1 Uniform choice for every $N \geq 3$

Increase C_1 to at least one. Take w_0 so small that $\frac{1}{4} \log_2(1/w) \geq 2$ for $w \leq w_0$, and set

$$K = \min \left\{ \left\lfloor \frac{N-1}{2} \right\rfloor, \left\lfloor \frac{1}{4} \log_2(1/w) \right\rfloor \right\}, \quad h = \left(\frac{w}{C_1 K^{1/(2s)}} \right)^{2/3}.$$

Then $K \geq 1$, $2K + 1 \leq N$, and $K \geq c \min\{N, \log(e/w)\}$ uniformly: the first floor is at least $N/4$ for $N \geq 3$, and the other floor is comparable to $\log(e/w)$ on this range. Also

$$h 2^K \leq C_1^{-2/3} w^{2/3-1/4} K^{-1/(3s)} \leq C_1^{-2/3} w^{5/12}.$$

One further decrease of w_0 , independent of N , therefore ensures the hypothesis of Lemma 4.1. Its target distance is at most w , and its map distance is at least $c w^{1/3} K^{1/(3s)}$, proving the desired lower bound. For every fixed N , K eventually equals $\lfloor (N-1)/2 \rfloor$ and stays fixed as $w \downarrow 0$; only h varies. There is no exchange of fixed and growing atom budgets.

4.2 A separate two-atom example

For $N = 2$ the preceding K would be zero, so use 008's rare-cell construction. For small $a > 0$, let $m_a = F(a)$ and choose $c_a \in (1 - k, 1 + k)$ with $\mathbb{E}F(a(c_a + kZ)) = F(a)$. Strict monotonicity and endpoint bracketing prove existence and uniqueness. Since $F(t) \sim ct^s$ at zero,

$$c_a \longrightarrow c_0, \quad \mathbb{E}(c_0 + kZ)^s = 1,$$

and, for $A_a = \{X < a\}$ and $C_a = \{X < a(c_a + kZ)\}$,

$$\rho(A_a \triangle C_a)/m_a \longrightarrow \mathbb{E}|(c_0 + kZ)^s - 1| > 0.$$

The last positivity uses nondegeneracy of Z . Put $R_a = m_a^{-1/p}$, $b_a = ka$ and $v_a = (-e_1 + b_a e_2)/\sqrt{1 + b_a^2}$. The two target probabilities

$$\mu_a = (1 - m_a)\delta_0 + m_a\delta_{-R_a e_1}, \quad \nu_a = (1 - m_a)\delta_0 + m_a\delta_{R_a v_a}$$

have p th moments exactly one, and are induced by the convex potentials $R_a(a - x_1)_+$ and $R_a(ac_a - x_1 + b_a x_2)_+/\sqrt{1 + b_a^2}$. Same-label coupling and the cell mismatch imply

$$W_2(\mu_a, \nu_a) \leq Cam_a^{1/2-1/p} \leq C'a^{3/2}, \quad \|T_{\mu_a} - T_{\nu_a}\|_2 \geq cm_a^{1/2-1/p} \geq c'a^{1/2},$$

using $m_a \asymp a^s$ and $1/2 - 1/p = 1/(2s)$. Choose $a = (w/C')^{2/3}$ and decrease w_0 once. This proves the $N = 2$ lower bound, since $\min\{2, \log(e/w)\}^\alpha$ is a fixed bounded factor on this range.

4.3 One atom

For point targets δ_y, δ_z with $|y|, |z| \leq 1$, both transport maps are constant and their $L^2(\rho)$ distance equals $|y - z| = W_2(\delta_y, \delta_z)$. Taking $y = -te_1/2$, $z = te_1/2$ with $t = \min\{w, 2\}$ attains the upper bound and respects both moment constraints. This proves the exact one-atom assertion and completes Theorem 1.1.

5 Interpretation and scope

For each $N \geq 2$, Theorem 1.1 gives

$$cN^\alpha \leq \liminf_{w \downarrow 0} \frac{\Omega_N(w)}{w^{1/3}} \leq \limsup_{w \downarrow 0} \frac{\Omega_N(w)}{w^{1/3}} \leq CN^\alpha.$$

No limit or exact leading constant is asserted. The logarithmic loss of the unrestricted problem saturates once the available number of atom levels is reached. The global density root still fails to belong to $W^{1,s}$: its derivative's s th power behaves as $1/x$ near the first boundary. Hence that strong root condition is not necessary for one-third stability on a fixed finite-atom class. This does not contradict its characterization of linear strong- L^s overlap, nor supply an arbitrary-source converse.

The constants and distance threshold are independent of N , but may depend on p, d and the fixed source. Vanishing atom weights as $w \rightarrow 0$ are permitted; a lower-weight constraint changes the question. Dimension one remains a separate isometric setting. No statement for general slowly varying boundary factors is made here.

The proof reuses 008's source and lower constructions, and (P) from 001; the new statement is their explicit finite-complexity continuation. The nearby semi-discrete results of Bansil–Kitagawa fix a common finite target support and control Laguerre-cell changes under weight perturbations, among other conclusions. Here positions and weights both vary under a moment constraint, and the metric is W_2 on targets versus L^2 on maps. This focused comparison does not certify historical originality.

References

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